

1. Prove that ${}_k|q_x = {}_kp_x q_{x+k}$
2. Z is the present value random variable for a 15-year pure endowment of 1 on (x):
 - (i) The force of mortality is constant over the 15-year period.
 - (ii) $v = 0.9$
 - (iii) $\text{Var}(Z) = 0.065E[Z]$

Calculate q_x

3. For a whole life insurance of 1000 on (x) with benefits payable at the moment of death:

- (i) The force of interest at time t, $\delta_t = \begin{cases} 0.04, & 0 < t \leq 10 \\ 0.05, & 10 < t \end{cases}$
- (ii) $\mu_{x+t} = \begin{cases} 0.06, & 0 < t \leq 10 \\ 0.07, & 10 < t \end{cases}$

Calculate the single benefit premium for this insurance.

4. For a special whole life insurance on (x), payable at the moment of death:

- (i) $\mu_{x+t} = 0.05, \quad t > 0$
- (ii) $\mu = 0.08$
- (iii) The death benefit at time t is $b_t = e^{0.06t}, t > 0$.
- (iv) Z is the present value random variable for this insurance at issue.

Calculate $\text{Var}(Z)$.

5. For (x):

- (i) K is the curtate future lifetime random variable
- (ii) $q_{x+k} = 0.1(k+1), \quad k = 0, 1, 2, \dots, 9$
- (iii) $X = \min(K, 3)$

Calculate $\text{Var}(X)$

6. For a special 3-year term insurance on (x), you are given:

- (i) Z is the present-value random variable for the death benefits.
- (ii) $q_{x+k} = 0.02(k+1) \quad k = 0, 1, 2$
- (iii) The following death benefits, payable at the end of the year of death:
- (iv) $i = 0.06$

7. For a special fully discrete 3-year term insurance on (x):

k	b_{k+1}
0	300,000
1	350,000
2	400,000

- (i) The death benefit payable at the end of year $k+1$ is

$$b_{k+x} = \begin{cases} 0 & \text{for } k = 0 \\ 1000(11 - k) & \text{for } k = 1, 2 \end{cases}$$

- (ii) The mortality probabilities are given as:

k	q_{k+1}
0	0.200
1	0.100
2	0.097

- (iii) $i = 0.06$

Calculate the level annual benefit premium for this insurance.

8. Based on a 4% annual interest rate, evaluate

(a) ${}_5E_{40}$

(b) D_{35}

(c) $A_{25:\overline{10}|}$

9. A maintenance contract on a hotel promises to replace burned out light bulbs at the end of each year for three years. The hotel has 10,000 light bulbs. The light bulbs are all new. If a replacement bulb burns out, it too will be replaced with a new bulb.

You are given:

- (i) For new light bulbs, $q_0 = 0.10$, $q_1 = 0.30$, $q_2 = 0.50$

- (ii) Each light bulb costs 1.

- (iii) $i = 0.05$

Calculate the actuarial present value of this contract.

10. A whole life insurance of 1 issued to (20) is payable at the moment of death. You are given:

- (i) The time-at-death random variable is exponential with $\mu = 0.02$

- (ii) $\delta = 0.14$

Calculate $\bar{A}_{20}$, ${}^2\bar{A}_{20}$ and $Var(\bar{Z}_{20})$

11. You are given the following information:

- (i) A unit benefit n-year term life insurance policy.
 - (ii) The age-at-death random variable is uniform on $[0, \omega]$
 - (iii) The constant force of interest δ
 - (iv) $\bar{Z}_{x:\overline{n}|}^1$ is the contingent payment random variable for a life aged x.
- Find $\bar{A}_{x:\overline{n}|}^1$, ${}^2\bar{A}_{x:\overline{n}|}^1$ and $Var(\bar{Z}_{x:\overline{n}|}^1)$

12. The age-at-death random variable is exponential with parameter 0.05. A life aged 35 buys a 25-year life insurance policy that pays 1 upon death. Assume an annual effective interest rate of 7%, find the actuarial present value of this policy.
13. The lifetime of a group of people has the following survival function associated with it: $s(x) = 1 - x/100$, $0 \leq x \leq 100$. Paul, a member of the group, is currently 40 years old and has a 15-year endowment insurance policy, which will pay him \$50,000 upon death. Find the actuarial present value and standard deviation of this policy. Assume an annual force of interest $\delta = 0.05$
14. You are given the following survival function for a newborn:

$$S_0(t) = \frac{(121-t)^{1/2}}{k}, \quad t \in [0, \omega]$$
 - (a) Show that k must be 11 for $S_0(t)$ to be a valid survival function.
 - (b) Show that the limiting age, ω , for this survival model is 121.
 - (c) Calculate e_5° for this survival model.
 - (d) Derive an expression for μ_x for this survival model, simplifying the expression as much as possible.
 - (e) Calculate the probability, using the above survival model, that (57) dies between the ages of 84 and 100.
15. Let the remaining lifetime at birth random variable be uniform on $[0, 250]$. let Z be the contingent payment random variable for a life aged x=30. Find $A_{30:\overline{10}|}^1$, ${}^2A_{30:\overline{10}|}^1$ and $Var(Z)$ if $\delta = 0.05$
16. Let the remaining lifetime at birth random variable be uniform on $[0, 100]$. let Z_{30} be the contingent payment random variable for a life aged x=30. Find A_{30} , ${}^2A_{30}$ and $Var(Z_{30})$
17. Let the remaining lifetime at birth random variable be uniform on $[0, 120]$. let ${}_{10|}Z_{30}$ be the contingent payment random variable for a life aged x=30. Find ${}_{10|}A_{30}$, ${}_{10|}{}^2A_{30}$ and $Var({}_{10|}Z_{30})$
18. Let the remaining lifetime at birth random variable X be exponential with $\mu = 0.06$. Let $Z_{30:\overline{10}|}^1$ be the contingent payment random variable for a life aged x=30. Find $A_{30:\overline{10}|}^1$, ${}^2A_{30:\overline{10}|}^1$ and $Var(Z_{30:\overline{10}|}^1)$ if $\delta = 0.10$

19. Let the remaining lifetime at birth random variable X be exponential with $\mu = 0.06$. Let $Z_{30:\overline{10}|}$ be the contingent payment random variable for a life aged $x=30$. Find $A_{30:\overline{10}|}$, ${}^2A_{30:\overline{10}|}$ and $Var(Z_{30:\overline{10}|})$ if $\delta = 0.10$